

- Q.25 Explain the race around condition in JK flip flop
- Q.26 Explain De Morgans's second theorem.
- Q.27 List five characteristics of TTL logic family.
- Q.28 Explain the working of 1:4 DEMUX.
- Q.29 What are the advantages of 2's complement method of subtraction over 1's complement.
- Q.30 Differentiate between Synchronous and Asynchronous counters.
- Q.31 Explain the working of R-2R ladder DAC
- Q.32 Explain the working of successive approximation A/D converter.
- Q.33 Differentiate among Mealy and Moore models
- Q.34 Explain the working of 3-bit ring counter.
- Q.35 Explain the working of universal shift register

SECTION-D

- Note:** Long answer type questions. Attempt any two questions out of three questions. (2x10=20)
- Q.36 Simplify using K-map and realize using only AND & OR gate $Y = \sum m(0,4,7,8,11,15)$ plus $d(1,5,6,9,10,14)$
- Q.37 Show that all the other gates can be obtained using only NAND gate
- Q.38 Explain with diagram the working of Decade counter.

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SECTION-A

Note: Multiple choice questions. All questions are compulsory (10x1=10)

- Q.1 Output of JK flip flop when both inputs are one is _____
a) 0 b) 1
c) Not defined d) Toggle
- Q.2 ASCII stands for?
a) American Stable Code for Interchange Information
b) American Standard Code for Interchange Information
c) American Standard Code for Information Interchange
d) American Stable Code for Information Interchange
- Q.3 Which logic gate is used to invert a signal?
a) AND gate b) OR gate
c) NOT gate d) XOR gate
- Q.4 3-input AND gate will give an output of 1 when:
a) All three inputs are 0
b) At least one input is 0
c) All three inputs are 1
d) At least one inputs is 1

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- Q.5 $(17)_8 = (\text{_____})_2$
 a) 00010001 b) 00010111
 c) 00001111 d) 11110000
- Q.6 A full adder adds
 a) Two bits
 b) Two bits and a carry
 c) Two 8-bit numbers and a carry
 d) Two 8-bit numbers and a carry
- Q.7 Number of flip-flops required to design a 4-bit binary counter is _____
 a) 3 b) 4
 c) 5 d) 6
- Q.8 Which of the following is a characteristics of CMOS logic?
 a) High power consumption
 b) Low power consumption
 c) Requires a high operating voltage
 d) Only operates at high frequencies
- Q.9 Which of the following is the correct standard representation for a 4-bit binary counter?
 a) 0001,0010,0011,0100
 b) 0000,0001,0011,0101
 c) 0000,0001,0010,0011
 d) 0001,0010,0011,0101
- Q.10 Number of select lines in 4 to 1 MUX is _____.
 a) 1 b) 2
 c) 3 d) 4

SECTION-B

Note: Objective type questions. All questions are compulsory. (10x1=10)

- Q.11 The decimal equivalent of the binary number 111 is _____
- Q.12 The base of the octal number system is _____
- Q.13 Expand TTL
- Q.14 Number of select lines in 2:1 MUX is _____
- Q.15 Full form of PIPO is _____
- Q.16 Full form of POS is _____
- Q.17 Output of SR flip flop when both the inputs are one is _____
- Q.18 A decade counter require _____ number of flip flops.
- Q.19 Ring counter is a combinational circuit. (True/False)
- Q.20 A half adder has _____ number of inputs.

SECTION-C

Note: Short answer type questions. Attempt any twelve questions out of fifteen questions. (12x5=60)

- Q.21 Differentiate between analog and digital signals.
- Q.22 Minimize using boolean algebra $\bar{A}\bar{B}C + \bar{A}B\bar{C} + A\bar{B}\bar{C} + A\bar{B}C + A\bar{B}C + ABC$
- Q.23 Convert $(123)_{16}$ into decimal and binary number
- Q.24 Draw the symbol and truth table of Ex-Not gate